ATR 72-600 SYSTEM MANUAL

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Operator context	Caribbean Airlines regional fleet
Purpose	Virtual-airline and flight-simulation training

CRITICAL NOTICE: This independently prepared manual is not an approved Caribbean Airlines, ATR, Pratt & Whitney Canada, EASA or TTCAA publication. It must never be used to operate, dispatch or maintain a real aircraft. Current aircraft-specific FCOM, QRH, AFM, SOP and performance data always prevail.

Document Control and Use

Purpose

ATR 72-600 SYSTEM MANUAL

This manual gives virtual flight crew a structured operating framework for the ATR 72-600 in Caribbean Airlines-style regional service. It explains the aircraft, crew coordination, flight-deck preparation, powerplant and propeller management, FMS setup, performance discipline, normal flight sequence and non-normal decision-making.

Caribbean Airlines Configuration Context

Caribbean Airlines identifies the ATR 72-600 as its principal intra-Caribbean and South American regional aircraft and publicly describes a 68-seat Economy Class layout. Individual registrations, equipment options, weights, hotel-mode authorization and avionics software can differ; the simulator model and current operator documents control.

Source Hierarchy

1	Current aircraft-specific AFM/FCOM/QRH, limitations and performance tool.
2	Caribbean Airlines SOP, MEL/CDL, dispatch release and training notices.
3	Current charts, navigation database, weather, NOTAMs and ATC clearance.
4	Simulator developer documentation and modeled-aircraft configuration.
5	This manual as explanatory training guidance.

CHECKLIST DISCIPLINE: Only the current aircraft-specific checklist supplies exact memory items, switch positions, engine limits and performance data. Generic flows in this manual must be verified against the modeled ATR.

Crew Abbreviations

PF	Pilot Flying — flight path, configuration requests and approach/landing.
PM	Pilot Monitoring — systems, communications, checklists and cross-checks.
CM1 / CM2	Captain / First Officer crew station references.
MCDU	Multipurpose Control and Display Unit.
PMS	Power Management Selector.

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1. General Information and Operating Philosophy

Scope

The ATR 72-600 is optimized for short regional sectors, frequent cycles and airports where runway length, pavement strength or infrastructure may limit larger jets. Its turboprop efficiency and low-speed performance suit the Caribbean network, but propeller, icing, wind and runway-condition management demand deliberate crew technique.

Operating Priorities

Aviate	Control attitude, speed, power and flight path; use the flight director only when credible.
Navigate	Protect terrain clearance, verify position and confirm the intended route.
Communicate	Coordinate with the other pilot, ATC, cabin crew and dispatch.
Manage	Complete checklists, monitor energy/fuel and make timely diversion decisions.

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Threat and Error Management

- Short sectors compress workload: prepare the arrival before departure or early in climb.
- Convective weather can isolate island destinations quickly; preserve diversion fuel and escape options.
- Wet runways, standing water and gusty crosswinds require current performance data.
- High propeller energy and ground clearance require strict ramp awareness.
- Automation does not correct a wrong FMS route, target altitude or power-management selection.

Stabilized Approach Policy

By 1,000 ft above airport elevation in IMC and 500 ft in VMC, the aircraft should be on the correct lateral and vertical path, in landing configuration, checklist complete, with target speed and stable power. Unless an approved procedure states otherwise, speed should normally remain within +10/−5 kt of target and sink rate should not exceed 1,000 ft/min without briefing. If not stabilized, go around.

2. Aircraft Description and Principal Data

Public Aircraft Baseline

ICAO type designator	AT76
Engines	2 × Pratt & Whitney Canada PW127M turboprops
Propellers	Six-blade constant-speed, full-feathering, reversible propellers
Caribbean Airlines seating	68 Economy Class passengers
Maximum takeoff weight	23,000 kg / 50,705 lb
Maximum landing weight	22,350 kg / 49,272 lb
Maximum zero-fuel weight	21,000 kg / 46,296 lb
Maximum fuel load	5,000 kg / 11,024 lb
Overall length	27.17 m / 89 ft 1.5 in
Wingspan	27.05 m / 88 ft 9 in
Range at maximum passengers	Approximately 740 NM / 1,370 km

Representative Operating Envelope

VERIFY VALUES: The following common ATR 72-600 training values are orientation data only. Confirm the exact aircraft placards and simulator limitations before flight.

Maximum operating altitude	25,000 ft
Maximum operating speed	250 KIAS
Maximum operating Mach	M0.55
Typical high-speed cruise	Approximately 270–275 KTAS
Takeoff distance at MTOW, ISA, sea level	Approximately 1,315 m / 4,314 ft
Landing distance at MLW, sea level	Approximately 915 m / 3,002 ft

Aircraft Geometry and Ground Risks

- The high wing provides good propeller clearance but places the wingtip and propeller discs outside the crew’s direct close-range view.
- The main passenger door and service/cargo access require ground-equipment coordination.
- Propeller blast can injure personnel or damage equipment; use minimum power and cleared areas.
- Tail clearance and nosewheel steering geometry require smooth rotation and disciplined taxi turns.

3. Flight Deck, Crew Duties and Standard Calls

Flight Deck Layout

Glareshield	Flight-guidance control panel, EFIS controls and warning attention.
Main panel	PFDs, MFDs, standby instruments, engine/warning display and landing gear.
Center pedestal	Power levers, condition levers, MCDUs, radios, flaps and parking brake.
Overhead panel	Electrical, fuel, hydraulics, pneumatics, anti-ice, air conditioning and lights.
Side consoles	Checklists, oxygen, tiller/steering and operator-installed equipment.

PF/PM Division

- PF briefs and controls the flight path, requests configuration, sets/monitors power and decides to continue or go around.
- PM completes flows, verifies modes and power, monitors speed/altitude, handles radios and reads checklists.
- Both pilots independently verify performance entries, FMS route, takeoff speeds, trim, minimums and fuel.
- Any crew member may call “STOP,” “UNSTABLE,” or “GO-AROUND.”

Mode and Power Awareness

ATR operation requires monitoring both autoflight modes and power-management state. Whenever a selection changes the aircraft’s strategy, the crew confirms the Flight Mode

Annunciator, Power Management Selector position, target torque and actual engine/propeller response.

4. Aircraft Systems Overview

Electrical	DC generators, AC wild-frequency sources, batteries and converters supply avionics/services.	Sources, buses, voltage/load and alert messages.
Hydraulic	Two electrically powered hydraulic systems serve flight controls and utility users.	Pump status, pressure/quantity and downstream capability.
Fuel	Wing tanks feed each engine through electric pumps and crossfeed logic.	Quantity, balance, pressure and planned burn.
Pneumatic	Engine bleed air supports air conditioning, pressurization, deicing and services.	Bleed configuration, temperatures and fault alerts.
Flight controls	Conventional powered/manual controls with spoilers, flaps and trim.	Control check, trim, configuration and warnings.
Avionics	Integrated displays, AFCS, FMS, radio navigation and alerting.	Valid sources, FMA, route and comparator flags.

Alerting Philosophy

The Flight Warning System prioritizes warnings, cautions and advisories through aural alerts, attention getters and system messages. The crew controls the aircraft first, identifies the message, performs memory actions only when required, and then reads the applicable checklist. Clearing a message does not correct the fault.

5. Electrical and Hydraulic Systems

Electrical Power

Each engine drives a DC starter-generator and an AC wild-frequency generator for associated loads. Batteries support engine start and essential buses. Static inverters/transformers and bus-tie logic provide required AC/DC distribution. External DC power supports ground operation when acceptable.

Power flow: battery / external / generator > main and essential buses > conversion/distribution > avionics, pumps and services

BAT switches	Connect batteries and enable normal charging/protection logic.
EXT PWR	Connects an acceptable ground source.
DC/AC generator controls	Connect or isolate generating sources.
Bus/voltage/load indications	Confirm power quality and loading, not just source availability.

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ELEC alert	May indicate a source, bus, converter or battery/charging problem.

Hydraulic Power

Hydraulic pressure is generated by electric pumps and distributed to flight-control and utility functions. Separate circuits and accumulators preserve selected capability following a pump or power failure. Because hydraulic pumps depend on electrical power, an electrical failure can produce secondary hydraulic indications.

Verify pump status before moving flight controls, flaps or gear.

Monitor quantity as well as pressure; falling quantity suggests leakage.

Do not cycle a failed or leaking system unless the checklist directs.

Use the QRH to identify retained braking, steering, gear and flight-control capability.

6. Fuel, Pneumatic, Air Conditioning and Pressurization

Fuel System

Integral wing tanks supply their respective engines through electric boost pumps. Crossfeed permits one side to supply both engines for approved balancing or failure procedures. Low-pressure and imbalance indications require prompt configuration review and leak assessment.

Fuel quantity	Compare loaded fuel, totalizer, dispatch release and planned burn.
Fuel pumps	Confirm pressure and source; a switch ON does not prove output.
Crossfeed	Use only by procedure; verify the supplying tank and pump configuration.
Fuel imbalance	Check pump/crossfeed state and consider leakage before balancing.
Fuel temperature	Monitor cold-fuel margins where modeled and applicable.

Pneumatic and Air Conditioning

Engine bleed air supplies the environmental control system and pressurization. Pack and bleed configuration changes with engine start, single-engine operation, deicing and abnormal conditions. The crew monitors temperature, bleed status and related alerts instead of judging system health from cabin comfort alone.

Pressurization

Automatic controllers schedule cabin altitude using aircraft altitude and landing-elevation data. Pack inflow pressurizes the fuselage while outflow valves regulate cabin pressure. Monitor cabin altitude, rate and differential pressure through climb and descent. A cabin-altitude warning requires oxygen, checklist action and descent planning.

7. Powerplant, Propellers and Power Management

PW127M Engine

Each PW127M is a free-turbine turboprop: the gas generator drives a separate power turbine connected to the reduction gearbox and propeller. Engine indications normally include gas-generator speed (NH), propeller speed (NP), torque, inter-turbine temperature (ITT), oil pressure/temperature and fuel flow.

NH	Gas-generator speed; central to start monitoring and engine condition.
NP	Propeller rotational speed; governed by condition-lever and power-management logic.
Torque	Primary shaft-power indication and power-setting reference.
ITT	Thermal limit/start-quality indication.
Oil pressure / temperature	Lubrication and gearbox/engine health.
Fuel flow	Engine symmetry and fuel-progress monitoring.

Propeller System

Six-blade, constant-speed propellers use variable blade angle to deliver efficient takeoff, climb, cruise and reverse thrust. The governing system maintains selected NP in flight. Feathering reduces drag after shutdown or failure. Reverse/beta range is for ground operation only and must be entered with the aircraft firmly on the runway or taxi surface.

Power and Condition Levers

Power levers	Command engine power and, in ground range, propeller blade angle/reverse.
Condition levers — FUEL SO	Shut off engine fuel and command shutdown.
Condition levers — FTR	Feather/start-related position as specified by procedure.
Condition levers — AUTO	Normal automatic propeller speed/control logic.
Condition levers — 100% OVRD	Commands high propeller speed for specified abnormal/operational use.
Power Management Selector	TO, MCT, CLB and CRZ schedules target power/propeller logic.

Hotel Mode / Propeller Brake

If installed, modeled and operator-authorized, the propeller brake can stop the No. 2 propeller while its engine gas generator supplies ground electrical and pneumatic services. This creates strong system and personnel considerations. Use only the exact ground procedure; confirm the propeller-brake indication and remain within time/temperature limits. Hotel mode is not an airborne operating mode.

PROPELLER SAFETY: Never move a condition or power lever based only on memory while people or equipment are near a propeller. Verify the intended engine, lever and ground clearance before action.

8. Automatic Flight, Navigation and Communications

Automatic Flight Control System

The AFCS combines flight directors, autopilot and yaw control with FMS or selected guidance. The Flight Guidance Control Panel supplies heading, altitude, speed and vertical targets. The FMA is the controlling indication of active and armed modes.

HDG / NAV	Selected heading or FMS lateral path.
ALT / VS / IAS	Selected altitude, vertical speed or speed-based climb/descent.
APP	Approach capture and tracking when correctly armed and supported.
GA	Go-around guidance after the applicable command.
Autopilot	Controls the flight path only within system engagement conditions.

Navigation and Surveillance

- FMS position blends inertial/GNSS and radio inputs according to the installed suite.
- Weather radar displays precipitation reflectivity; attenuation can hide hazards beyond strong cells.
- TAWS/EGPWS warnings demand immediate terrain-escape response.
- TCAS resolution advisories take priority over ATC vertical instructions unless aircraft safety requires otherwise.
- File only equipment and surveillance codes supported by the aircraft and network client.

9. FMS/MCDU Programming

Initialization

1. Apply stable electrical power and verify avionics/display availability.
2. Confirm aircraft type, navigation database validity and current position.
3. Enter origin, destination, flight number and company route if applicable.
4. Build the route from the clearance; do not delete a discontinuity until continuity is intended.
5. Select departure, arrival and approach procedures from current charts.

Enter weight, fuel, cruise altitude, wind and performance data as supported.

Enter or verify takeoff runway, flap, speeds and thrust/power data from the performance source.

Step through every leg on the map and compare constraints with the operational flight plan.

Route Verification

Check for duplicate fixes, course reversals, impossible turns and route gaps.

Confirm initial climb altitude and missed-approach altitude against clearance.

Verify the destination runway, approach transition and minima.

Compare FMS distance/fuel prediction against dispatch values.

On short sectors, complete arrival setup before top of climb whenever practical.

Vertical Navigation Discipline

The FMS provides predictions and guidance but cannot compensate for an incorrect wind, weight, route or constraint. Monitor vertical deviation, selected modes and actual energy. If high, reduce energy early using speed and drag within limits; do not chase the path close to terrain or the stabilization gate.

10. Flight Planning, Fuel and Weight & Balance

Fuel Components

Taxi	Engine/hotel-mode/APU use, start, taxi and expected delay.
Trip	Takeoff to landing for forecast route, level, wind and performance.
Contingency	Uncertainty in wind, routing, level and aircraft performance.
Alternate	Missed approach, route to alternate and landing.
Final reserve	Protected minimum reserve; not normal discretionary fuel.
Additional / extra	Weather, holding, remote destination, MEL/CDL or captain’s margin.

Weight and Balance

Confirm passenger, baggage, cargo and fuel inputs.

Verify zero-fuel weight and center of gravity against the loadsheet.

Confirm ramp, takeoff and predicted landing weights are within limits.

Enter weight and CG data consistently in the simulator, FMS and performance tool.

Set takeoff trim from the approved source and cross-check before takeoff.

Island-Alternate Planning

A geographically close island is not automatically a suitable alternate. Assess runway length/condition, forecast and observed weather, navigation aids, rescue capability, opening hours, handling, fuel and the route between airports. Preserve enough fuel to reach the selected alternate after a missed approach under credible weather deviation.

11. Takeoff and Landing Performance

Takeoff Calculation

- Use a valid performance tool for runway, intersection, slope, wind, temperature, pressure, surface condition and obstacle data.
- Confirm takeoff weight, flap setting, engine/propeller configuration, bleed/deicing assumptions and runway.
- Cross-check V1, VR and V2 for order and reasonableness.
- Brief acceleration altitude, engine-out flight path, terrain, return runway and immediate diversion options.
- Recalculate after material changes to runway, wind, temperature, weight or contamination.

Landing Calculation

- Use actual landing weight, runway condition, wind, slope and planned flap/reverse/braking configuration.
- Compare required distance with landing distance available using operator safety factors.
- Account for tailwind, gusts, standing water, inoperative equipment and anticipated touchdown point.
- A comfortable touchdown must not be achieved by floating beyond the touchdown zone.
- Go around if runway occupancy, wind, braking or approach stability invalidates the plan.

Reference Speeds

VAPP is normally built from the applicable reference speed plus operator wind correction. Use the aircraft-specific calculation; do not invent a large additive in gusty conditions because excessive speed increases landing distance. Respect flap, gear and structural speed limits displayed or published for the modeled aircraft.

12. Preliminary Preflight and Cockpit Preparation

Dispatch Review

- Review technical log, MEL/CDL, dispatch release, weather, NOTAMs, route and alternates.

Check fuel plan, payload, runway performance and operational limits.

Identify major threats and define delay, diversion and no-go triggers.

Confirm charts/navigation database and simulator aircraft version.

Exterior Inspection

General condition, leaks, panels, doors, static ports and probes.

Propeller blades/spinners, engine inlets/exhausts and visible damage.

Landing gear, tires, brakes, hydraulic lines and wheel wells.

Wing/empennage surfaces, ice/contamination, lights and control surfaces.

Fuel caps/vents, drain evidence and cargo/passenger door security.

Ground equipment, chocks and personnel clear of propeller arcs.

Cockpit Preparation Flow

Overhead	Batteries/external power, fire test, lights, fuel, hydraulics, anti-ice, bleeds and packs.
Main panel	Displays, standby instruments, altimeters, landing gear, brakes and warnings.
Pedestal	Power/condition levers, flaps, parking brake, MCDUs, radios and transponder.
Guidance panel	Initial speed, heading, altitude, flight directors and modes.
Cabin/doors	Passenger signs, cabin coordination, doors and loadsheet status.

13. Before Start and Engine Start

Before Start

Loadsheet accepted; fuel, weight, CG, trim and performance agree across sources.

Departure briefing complete: taxi, runway, weather, speeds, power, engine-out path and return strategy.

Doors closed, passengers seated, ground crew communications established.

Beacon/propeller-area warning active before engine movement.

Power and condition levers in the aircraft-specific starting positions.

Engine Start Monitoring

Confirm adequate battery/external power and cleared propeller area.

Select the intended engine start command in accordance with the modeled sequence.

Verify gas-generator rotation and oil-pressure response.

Introduce fuel with the correct condition-lever action at the specified NH.

Monitor light-off, ITT, NH, NP, oil pressure and starter disengagement.

Abort for no light-off, hot start, hung start, no oil pressure or abnormal noise/indication.

Stabilize the engine, connect/verify generating sources and reconfigure bleeds/packs.

Start the second engine and complete the post-start flow.

Hotel-Mode Departure Consideration

If No. 2 was operating in hotel mode, removal of the propeller brake and transition to normal propeller operation must follow the precise ground procedure. Confirm the area is clear, brake disengagement is indicated, NP is normal, and cabin/ground personnel understand that the propeller will begin turning.

14. Taxi and Before Takeoff

Taxi Technique

Release brakes only after confirming the route and clearance.

Complete an early brake check and verify steering response.

Use minimum power; avoid excessive beta/reverse and propeller blast.

Reduce speed before turns, wet areas, congestion and uneven pavement.

Maintain wingtip and propeller-disc clearance from vehicles, buildings and personnel.

Flight-Control Check

In a clear area, move controls smoothly through the required range while PM confirms correct surface/indication response and freedom of movement. Confirm trim, flaps and spoilers. Avoid prolonged or abrupt checks in strong wind.

Before Takeoff Configuration

Flaps	Performance-selected setting and correct indication.
Trim	Set to loadsheet/performance value and cross-checked.
PMS	Takeoff position required by performance data.
Condition levers	Takeoff/automatic position according to SOP.
Deicing / anti-ice	Configuration matches conditions and performance calculation.
Takeoff speeds	Correct for runway, weight and configuration.
Flight guidance	Correct lateral/vertical plan and cleared altitude.
Cabin	Secured and crew advised.

15. Takeoff and Initial Climb

Takeoff Sequence

Enter the runway only when cleared and confirm runway/heading.

Align smoothly; hold brakes if required and stabilize power according to SOP.

Set takeoff power using the prescribed power-lever/PMS technique.

PM verifies target torque, engine symmetry and no limiting exceedance.

Call 80 knots, V1 and Rotate using operator callout policy.

Rotate smoothly to the target attitude; follow valid flight-director guidance.

After positive climb, retract landing gear.

At acceleration altitude, lower pitch, accelerate and retract flaps on schedule.

Engine Failure Awareness

At high power and low speed, an engine failure produces yaw and loss of climb performance. Maintain directional control, set the required attitude, verify positive climb and retract gear. Do not rush identification. At a safe altitude, identify and confirm the failed engine, complete memory actions/checklist, then plan a suitable landing.

After Takeoff

Set climb power through the correct PMS and lever logic.

Confirm flaps retracted, gear up and systems normal.

Engage automation only after confirming valid modes and stable flight path.

Complete the After Takeoff checklist and compare fuel/time with plan.

16. Climb, Cruise and Descent

Climb

Monitor torque, ITT, NP/NH, oil and engine symmetry.

Use target IAS and climb power appropriate to weight, icing and terrain.

Cross-check pressurization and cabin rate early.

Set standard pressure at transition altitude and verify both sides.

Anticipate reduced climb margin in icing, turbulence or high temperature.

Cruise

Level-off	PMS/lever state, cruise speed, fuel flow, altitude and FMS prediction.
Regular scan	Fuel balance/progress, engines, electrics, hydraulics, cabin and weather.

Before remote segment	Communications, suitable airports, weather escape and fuel.
Before descent	ATIS/weather, NOTAMs, landing performance, arrival and approach setup.

Descent Planning

A useful initial estimate is about 3 NM per 1,000 ft to lose, adjusted for deceleration, wind and restrictions. Begin early enough to avoid high power followed by aggressive drag. Monitor speed closely with icing equipment or strong tailwind. Complete landing data and approach briefing before workload peaks.

17. Approach, Landing and Go-Around

Approach Setup

- Verify procedure, navigation source, minima, missed approach and terrain.
- Confirm landing weight, flap, VAPP, wind correction and landing distance.
- Brief power/automation plan, visual references, runway exit and go-around triggers.
- Set pressurization landing elevation and complete cabin coordination.

Configuration Strategy

Configure early enough to stabilize without excessive low-altitude power changes. Respect flap and gear extension limits. Select landing gear when required for drag and complete the Landing checklist by the stabilization gate. Monitor NP, torque, speed trend and wind correction throughout final approach.

Landing Technique

1. Maintain target speed and aim point; correct drift with coordinated control.
2. At the appropriate height, transition to a modest flare without floating.
3. Reduce power smoothly toward flight idle as required by the modeled technique.
4. Touch down in the touchdown zone with the aircraft aligned.
5. After weight-on-wheels, use ground range/reverse only as authorized and monitor directional control.
6. Apply braking appropriate to runway condition and planned exit.

Go-Around

1. Apply go-around power and call the maneuver.
2. Set the required pitch/flight-director guidance and control yaw.

Select the go-around flap setting as requested and verify climb.

With positive rate, retract gear.

Follow missed-approach track/altitude; accelerate and retract flaps on schedule.

Complete checklist, communicate and decide whether to re-approach or divert.

18. After Landing, Shutdown and Secure

After Landing

Clear the runway completely before heads-down actions.

Retract flaps, reset trim/spoilers as required and configure lights/transponder.

Set propeller/condition and PMS configuration for taxi.

Start or select the planned ground power source before shutdown.

Use minimal reverse/beta and protect personnel from propeller blast.

Shutdown

Park on stand and coordinate parking brake/chocks.

Establish external power or approved hotel-mode/ground power configuration.

Complete engine cooldown as required.

Move condition levers to the aircraft-specific shutdown position and monitor rundown.

Switch off fuel pumps, generators/bleeds and associated systems according to the flow.

Turn off the beacon only when propellers have stopped and it is safe for ground personnel.

Secure

Parking brake/chocks confirmed.

Batteries, external power and avionics configured for termination.

Fuel, hydraulics, bleeds/packs, anti-ice and lights off as required.

Doors and emergency lighting coordinated with cabin crew.

Technical discrepancies recorded before leaving the aircraft.

19. Adverse Weather and Icing

Icing Operations

The ATR’s pneumatic deicing boots and electrically heated surfaces protect critical areas, but they do not make prolonged severe icing safe. Use anti-ice/deicing according to published criteria, monitor ice evidence and performance, maintain the required speeds, and exit conditions when accretion or system capability becomes doubtful.

- Use engine intake anti-ice and propeller heat as prescribed.
- Activate airframe deicing through the correct automatic/manual logic.
- Respect icing speed additives and configuration restrictions.
- Expect increased drag, reduced climb performance and higher power.
- Do not delay escape after severe icing indications or unexpected performance loss.

Thunderstorms and Heavy Rain

- Avoid cells laterally; weather radar cannot guarantee a safe gap.
- Use tilt/range actively and remember attenuation behind strong returns.
- Expect windshear, rapid runway changes and unreliable braking reports.
- Hold away from cells or divert before fuel becomes critical.

Turbulence and Strong Wind

Use the published turbulence speed, seat-belt/cabin coordination and smooth control inputs. In gusty crosswinds, preserve speed discipline without carrying excessive additive. If control, alignment or touchdown-zone criteria are not assured, go around.

20. Caribbean and Special Airport Operations

Regional Threats

Fast-moving convection	Continuous weather review, escape route and early diversion decision.
Short/wet runway	Current performance, conservative touchdown point and braking plan.
Limited alternates	Protect alternate fuel and verify airport availability/services.
Mountainous islands	Brief terrain, minimum altitudes, missed approach and visual illusions.
Salt/humidity/heavy rain	Expect visibility, braking and maintenance effects; report discrepancies.
Bird activity	Use lights/SOP, review reported activity and remain prepared to reject/go around.

Short Sector Workload

On sectors such as Trinidad–Tobago or closely spaced island services, the arrival can begin soon after takeoff. Enter and brief the likely arrival before departure when practical, but never allow anticipated clearance to replace the actual ATC clearance. Sterile-cockpit discipline is essential.

Remote or Non-Radar Operations

- Confirm position-reporting and communication capability.
- Review minimum safe altitudes and terrain escape options.
- Carry fuel for credible routing and weather uncertainty.
- Verify destination opening hours, lighting, rescue and ground handling.

21. Non-Normal and Emergency Framework

CONTROLLING SOURCE: Memory items and exact checklist actions must come from the current ATR QRH/ECL. The following material describes priorities and decision logic only.

Universal Response

1. Maintain control and establish a safe flight path.
2. Identify the condition using alerts, indications and crew cross-check.
3. Complete only genuine memory actions.
4. Read and complete the applicable checklist.
5. Assess landing urgency, weather, terrain, runway, fuel and support.
6. Communicate with ATC, cabin crew and dispatch; brief approach and evacuation risk.

Rejected Takeoff

At low speed, reject for significant abnormality or doubt. Near V1, reject only for high-severity events defined by SOP, such as fire, engine failure, unsafe configuration or aircraft unable to fly. At or after V1, continue unless the aircraft is clearly incapable of flight. Maintain directional control and use reverse/braking deliberately.

Engine Fire or Failure

- Control yaw, speed and flight path.
- Confirm positive climb before gear retraction.
- Identify and confirm the affected engine; avoid wrong-lever action.
- Complete checklist actions at a safe altitude unless fire/severe damage demands immediate action.
- Plan a prompt landing for fire, severe damage or persistent abnormal indications.

Smoke, Pressurization and Unreliable Indications

- Smoke/fire/fumes: protect crew, use checklist and land at the nearest suitable airport without unnecessary troubleshooting.
- Cabin altitude: don oxygen, communicate, descend and divert according to the checklist.

- Unreliable airspeed: use known pitch/power, compare independent sources and reduce automation.
- Electrical/hydraulic fault cascades: search for the common upstream source and preserve remaining capability.

22. CRM, Training and Proficiency

Briefing Standard

A useful briefing states the plan, major threats, aircraft configuration, performance decisions and explicit triggers for reject, go-around or diversion. It is short enough to retain and ends by inviting challenge from the other pilot.

Initial Training Sequence

1 — Cockpit/systems	Power-up, panel scan, levers, PMS, displays and FMS.
2 — Ground/normal	Start, hotel mode if applicable, taxi, takeoff and normal landing.
3 — Instrument	AFCS modes, ILS/RNAV, missed approach and raw-data monitoring.
4 — Adverse weather	Icing, turbulence, windshear awareness and wet-runway decisions.
5 — Non-normal	Rejected takeoff, engine failure, fire, pressurization and diversion.
6 — Line-oriented	Complete Caribbean sector with realistic weather, fuel and ATC.

Proficiency Standard

- Maintains flight path and speed within training tolerance.
- Uses PMS, power/condition levers and automation with correct mode awareness.
- Completes FMS/performance setup and independent cross-checks.
- Recognizes unstable approach and goes around without prompting.
- Manages fuel, weather and alternates before margins become critical.
- Uses checklists without omitted safety-critical items.

Appendix A — Normal Checklist Reference

USE: This reference is a training checklist framework. Match wording and order to the exact Caribbean Airlines or simulator checklist before use.

Preflight	Oxygen • flight instruments • windows/doors • power/condition levers • parking brake
Before Start	Fuel/weight • FMS/performance • trim • briefing • doors • beacon

After Start	Generators • bleeds/packs • hydraulics • anti-ice • engine/propeller indications
Before Taxi	Brakes/steering • flight controls • flaps • trim • ground equipment clear
Before Takeoff	Cabin • runway • speeds • PMS • condition levers • flaps/trim • lights
After Takeoff	Gear • flaps • climb power • systems • altimeters
Descent	Landing data • weather • approach briefing • pressurization • fuel
Approach / Landing	Altimeters • cabin • gear • flaps • condition levers • checklist complete
After Landing	Flaps • trim • lights • transponder • anti-ice • power configuration
Shutdown / Secure	Parking/chocks • engines • pumps • hydraulics • bleeds • batteries • lights

Appendix B — Standard Callouts

Takeoff power	“Power set” after target torque and symmetry verified.
Acceleration	“80 knots” — cross-check airspeed.
Decision/rotation	“V1” / “Rotate.”
Initial climb	“Positive climb” / “Gear up.”
Altitude capture	“1,000 to go” and altitude capture/level call.
Approach	Mode captures, “Stable” or “Go-around.”
Minima	“Approaching minimums” / “Minimums.”
Landing	“Spoilers/reverse” and deceleration monitoring as applicable.
Abnormal	Alert name, flight path, checklist title and action confirmation.

Appendix C — Quick Flows and Decision Gates

Normal Flow

Power-up	Batteries / external power • fire test • displays • lights
Preflight	Fuel • hydraulics • bleeds/packs • anti-ice • FMS • exterior
Start	Area clear • source power • start • fuel • monitor • reconfigure
Taxi	Brake check • controls • flaps • trim • departure review
Takeoff	Runway verify • PMS/lever state • power set • calls • rotate
Climb/Cruise	Gear/flaps • climb power • pressurization • fuel/weather
Descent/Approach	Landing data • briefing • minima • configuration • stable gate
Landing/Shutdown	Touchdown zone • ground range • taxi flow • power • engines • secure

Go/No-Go Gates

Dispatch	Legal release, suitable weather/alternates, adequate fuel and serviceable aircraft.
Before start	Loadsheet/performance complete; doors and propeller area controlled.
Before takeoff	Correct runway, configuration, power data, clearance and no critical alert.
Approach gate	Correct path/configuration, checklist complete and stable speed/power/sink.
Landing commitment	Runway adequate/clear, aircraft controlled and touchdown zone assured.

Appendix D — Glossary and References

AFCS	Automatic Flight Control System
AFM	Aircraft Flight Manual
AT76	ICAO designator for ATR 72-600 family operations
ITT	Inter-Turbine Temperature
MCDU	Multipurpose Control and Display Unit
MEL / CDL	Minimum Equipment List / Configuration Deviation List
NH / NP	Gas-generator speed / propeller speed
PMS	Power Management Selector
QRH	Quick Reference Handbook
TAWS	Terrain Awareness and Warning System
VAPP	Approach target speed
V1 / VR / V2	Decision, rotation and takeoff safety speeds

Public References

ATR, ATR 72-600 aircraft specifications page, accessed August 2026.

Caribbean Airlines, official fleet/About Us information for ATR 72-600 seating and regional use.

EASA type-certificate and operational data applicable to the ATR 42/72 family.

Current simulator developer documentation, FCOM/QRH and model release notes.

FINAL REMINDER: This manual supports study and flight simulation. Use the current approved aircraft-specific manuals, checklists and performance data for any real operation.